Lesson notes · for you

Your connected project

IoT · Level 4 · Lesson 4

Materials
Paper, a pencil and a ruler; Two people who have the problem, to talk to; Optional: the Decide with numbers lab on a phone

Learning goals

- Turn a problem in your community into a connected design: sense, decide, act, connect.
- Weigh designs fairly against cost, power, safety and privacy.
- Present the design so others can judge and improve it.

Activity: Design a connected solution

1. Choose a real problem near you: a tank that runs dry, a hot classroom, a flooded path, a busy market. Ask two people who live with it what goes wrong, when, and who gets hurt.
2. Write one sentence you could check later, such as “no vehicle drives into water deeper than 15 cm” or “the tank never runs dry during school hours”.
3. For each design, draw four boxes joined by arrows: what it senses, the rule it decides by, what it does, and who it tells and how.
4. Make a table: score each design from 1 to 5 on cost, power, safety and privacy, multiply by the weights you agreed, and add up. Write what each design does when its sensor or battery fails.
5. Show your drawings and table to a friend or someone at home. Ask: what did I miss? Write down what they say, and one change you'd make.

Never test anything in floodwater, near mains electricity or on a road. Say who you are and why you're asking before you talk to people you don't know, and don't photograph people without asking.

Discussion

- Which design scored best, and would the people affected agree?
- What does your design do when its sensor fails?
- What data does your design collect, and does it need all of it?

Check yourself

- Name the four parts of a connected design.

Try it this week
Your connected project: pick a problem near you, draw your design as four boxes (sense, decide, act, connect), score two designs, then ask someone at home what you missed.

Worksheet

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Name _____ Class _____ Date _____

1 **Grace's design scored best, but she never asked the minibus drivers about it. What should she do before building it?**

- ◯ Build it now: the scores are clear
- ◯ Score it again herself, more carefully
- ◯ Show the drivers and ask what she missed

2 **Put the steps of a connected project in order.**

- ☐ Write a success measure you can check
- ☐ Sketch designs as sense, decide, act, connect
- ☐ Talk to the people who have the problem
- ☐ Present it and ask what you missed
- ☐ Score the designs with agreed weights

3 **Match each part of the bridge design to its job.**

A water-depth sensor on the bridge post	• • • •	Act
A small computer asking “deeper than 15 cm?”	• • • •	Connect
A flashing amber light by the road	• • • •	Sense
A text alert to drivers before they set off	• • • •	Decide

4 **Work out the light-only design's weighted score.**

Scores: cost 5, power 5, safety 2, privacy 5. Weights: cost 3, power 3, safety 5, privacy 4. Total: 15 + 15 + _____ + 20 = _____ out of 75, which is 80%.

Word bank: 7 · 10 · 65 · 60

5 **Which of these designs fail safe?**

- ☐ A heater that keeps heating if its thermometer breaks
- ☐ A railway crossing barrier that comes down if its power fails
- ☐ A flood light that flashes when it can't read its sensor
- ☐ A gate alarm that goes silent when its battery is low
- ☐ A tank pump that stops if its water-level sensor stops answering

✎ **Which problem near you would you most like to solve with a connected design? Who would you need to talk to first?**

Check your answers

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1 **Show the drivers and ask what she missed**

Scores only hold what Grace thought of, however carefully she scores.

2 **Top to bottom: 2, 3, 1, 5, 4**

Start with people and a goal you can check, so you know what you're designing for.

3 **A water-depth sensor on the bridge post — Sense**
A small computer asking "deeper than 15 cm?" — Decide
A flashing amber light by the road — Act
A text alert to drivers before they set off — Connect

The sensor measures, the computer applies the rule, the light acts on the spot, and the message reaches people who aren't there yet.

4 **Scores: cost 5, power 5, safety 2, privacy 5. Weights: cost 3, power 3, safety 5, privacy 4. Total: 15 + 15 + 10 + 20 = 60 out of 75, which is 80%.**

Safety scores 2 and weighs 5, so it adds 2 × 5 = 10 (adding 2 + 5 = 7 is the trap).

5 **A tank pump that stops if its water-level sensor stops answering; A flood light that flashes when it can't read its sensor; A railway crossing barrier that comes down if its power fails**

Failing safe means doing the harmless thing when something breaks: stop the pump, warn the drivers, close the crossing.